Predicting Art Prices at Auction

Machine Learning Project Report
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Contents

1. Abstract

Art valuation remains opaque, dominated by legacy auction houses and networks that restrict transparency and accessibility. This lack of independent verification of expected auction prices creates inefficiencies for artists, collectors, and investors. Existing approaches rely on expert judgment and historical trends, which are subjective, fragmented, and fail to scale. These methods do not provide predictive capabilities or actionable insights for stakeholders.
This project introduces a machine learning model that predicts future auction sale prices using features such as artist career stage, medium, condition, and market demand. Unlike traditional methods, the model leverages data-driven analysis to improve accuracy and accessibility, mitigating reliance on institutional gatekeeping. Preliminary results demonstrate weak performance, achieving R2 Score of -0.0509212947482387 and MAE of 3693.2598680948486 on a limited test dataset of historical auction data from a single auction house. This rationale behind this solution is to offer a scalable, transparent tool for auction preparation and performance assessment, with future iterations incorporating sentiment analysis to capture emotional drivers of art valuation.

2. Introduction

Art valuation remains a complex and opaque process, dominated by legacy auction houses and elite networks concentrated in major cities. This exclusivity limits transparency and creates barriers for artists, collectors, and investors who seek independent verification of expected auction prices. The inability to accurately predict sale outcomes introduces inefficiencies and risk, particularly for new entrants navigating the art market. Traditional valuation methods rely on expert judgment and historical precedent. While these approaches have shaped the industry for decades, they are inherently subjective, fragmented, and resistant to scale. They fail to provide predictive capabilities or leverage structured transaction data, leaving stakeholders dependent on institutional gatekeeping. As a result, the art market lacks accessible, data-driven tools that can support decision-making for auctions and investments.

This project addresses these limitations by proposing a machine learning model to predict future auction sale prices using quantifiable features such as artist career stage, medium, condition, authenticity, and market demand. Although emotional and sentiment-driven factors remain outside the scope of this initial work, future iterations may incorporate these dimensions. The project sourced a dataset of historical auction prices from a single auction house, conducted exploratory data analysis (EDA), and prepared the data by handling missing values and defining target and feature variables. The project then implemented a benchmark model for initial comparison, followed by a primary predictive model both of which used **CatBoost**, chosen for its ability to handle categorical variables effectively. The model was trained and tested using an 80-20 split. Preliminary results demonstrate weak performance, achieving an **R^2 score of -0.0509** and a **Mean Absolute Error (MAE) of 3,693.26** on a limited test dataset. These findings highlight the challenges of modelling art prices with limited data and underscore the need for broader datasets and feature enrichment to improve predictive accuracy.

3. Methodology

3.1. Methodology

This section outlines the steps and logic that the project followed.

1. Environment Setup and Imports
   - Import: numpy, pandas, matplotlib.pyplot, CatBoostRegressor, train_test_split, mean_absolute_error, r2_score.
2. Data Loading
   - Load a CSV file named artDataset.csv into a pandas DataFrame: data = pd.read_csv('artDataset.csv').
3. Exploration
   - Execute data.head()
   - Execute data.isnull()
4. Year Cleaning
   - Filter rows to keep only those where yearCreation is exactly 4 digits:
     i. data = data[data["yearCreation"].astype(str).str.match(r"^\d{4}$")]
   - Convert yearCreation to integer:
     i. data["yearCreation"] = data["yearCreation"].astype(int).
5. Price Cleaning
   - Change price column to strings.
   - Remove the literal substring " USD" and all "." characters.
   - Convert price to numeric with errors="coerce".
   - Drop rows where price conversion results in NaN.
6. Benchmark Model (CatBoost) Setup
   - Define a CatBoost regressor with: iterations=100, depth=5, learning_rate=0.05, loss_function='RMSE', verbose=100.
7. Benchmark Features and Split
   - Select features: X_benchmark = data[["yearCreation", "movement"]].
   - Target: y_benchmark = data["price"].
   - Split with train_size=0.8, random_state=26 to get X_train_b, X_test_b, y_train_b, y_test_b.
8. Benchmark Categorical Specification and Fit
   - Declare categorical features list: categorical_features = ["movement"].
   - Train with categorical features: benchmark.fit(X_train_b, y_train_b, cat_features=categorical_features).
9. Benchmark Prediction and Evaluation
   - Predict on X_test_b.
   - Print predictions.
   - Evaluate with $\mathbf{R^2}$ and **MAE** on test set; print both.
10. Primary Model (CatBoost) Setup
    - Define a CatBoost regressor with: iterations=1000, depth=10, learning_rate=0.05, loss_function='RMSE', verbose=100.
11. Primary Features and Split
    - Features: X = data.drop(columns=["price"]) (keep all columns except price).

- Target: y = data[”price”].
- Split with train_size=0.8, random_state=2025 to get X_train, X_test, y_train, y_test.

12. Primary Categorical Specification and Fit
    - Declare categorical features list: categorical_features = [”artist”, “title”, “signed”, “condition”, “period”, “movement”].
    - Train: cat_boost.fit(X_train, y_train, cat_features=categorical_features).
13. Primary Prediction and Evaluation
    - Predict on X_test.
    - Evaluate with $\mathbf{R^2}$ and **MAE** on test set; print both.

3.2. Baseline Model

The initially planned benchmark algorithm for the project was “Linear Regression” for a model trained only on year and price. However, going forward, the project will instead implement a simplified CatBoost that includes a subset of features (year, price, movement), with no parameter tuning.

3.3. Algorithms

a. CatBoost – A Gradient Boost Algorithm

Gradient Boost algorithms work by minimizing the loss function i.e. they work by identifying where things are inaccurate. This is done by creating sequential decision trees that correct the error with each new iteration, until the final model is as close to the target (accurate) as possible. Each new tree corrects the error from the previous tree and not the error of the entire problem. CatBoost improves on this by introducing ordered boosting and ordered target statistics. The algorithm has interfaces including *CatBoost* (a general model), *CatBoostRegressor* (for prediction problems) and *CatBoostClassifier* (for classification problems) (CatBoost, na).

How it works: gradient boosting -> ordered boosting -> categorical feature handling -> regularization

Features of the Algorithm:

- **Ordered Boosting –** GBM algorithms tend to be biased because they may predict on future data points that leak. CatBoost prevents this by using a permutation-driven approach to handle data.

- **Ordered target statistics –** CatBoost uses ordered target statistics, meaning that it shuffles the data and performs its tasks on the datapoints from the previous row and not the current row. This allows the algorithm to work as if it were dealing with data in real-time where it wouldn’t have upfront access to future data. In addition, the algorithm ensures that it uses only the models gradients and categorical features, not their target values. In other words, the algorithm looks at the error of the data point(s) and categorical features but does not peek at what the data point will predict.

- **Prevents over-fitting –** once the model detects overfitting based on the user-defined validation data, CatBoost can stop the training process.

- **Categorical features –** CatBoost works well with non-numerical features. The algorithm makes it possible to set/identify categorical (non-numerical) features which are then integrated to the learning process. To achieve this, the algorithm calculates a statistic for the feature based on previous rows. This ability to work with diverse features also reduces the need for extensive feature engineering since the algorithm can learn from more feature types.

- **Strong regularization –** regularization penalizes complex models to reduce their variance and improve their ability to generalize on unseen data. CatBoost applies regularization to models.

In summary, CatBoost applies the same approach to learning as Gradient Boost in the process of how both algorithms work. Their difference lies in how each algorithm handles the data it uses for training. Unlike regular GBM, CatBoost reduces bias by ensuring that there is no data leakage, meaning that current data is not used in predictions.

b. Limitations of CatBoost

Some limitations of CatBoost are:
- It is slightly slower than of GBM such as LightGBM
- It is potentially more computationally expensive because of the iterations
- It may not be relevant or work as well as other algorithms when there are no categorical features

3.4. Evaluation Metrics

The evaluation metrics chosen for the project are:
- R^2 – this explains how much variation in the target variable is explained by the model.
- Mean Absolute Error (MAE) – measures the average size of the prediction errors.

4. Results and Analysis

You should have a few tables/charts showing the results of your experiments. What was the best model? What were the best parameters? What tradeoffs did you find?

4.1. Results and Analysis

This model has not performed well in predicting future auction prices. Metrics on the benchmark model showed R2 Score: 0.1422346505224652 and MAE: 5063.975793354287. This low R2 Score suggests that the benchmark model performed really badly in predicting the mean. The high MAE (which is denoted in the target value: price) suggests that the model is quite far off in predicting price by a few thousand dollars. For example, if the average artwork price is $10,000, an MAE of $5,000 means the model is off by approximately 50% on average.

Metrics for the primary model showed R2 Score: -0.0509212947482387 and MAE: 3693.2598680948486. This low R2 Score suggests that the primary model performs worse than a simple baseline that predicts the aaverage price. The high MAE (which is denoted in the target value: price) suggests that the model is off in predicting price by a few thousand dollars. However, the MAE here was smaller than in the benchmark prediction suggesting a better performance.

4.2. Possible Reasons for this Performance

There are a few possible reasons why the model has performed like this:

- High cardinality across various features may likely have led to overfitting. Cardinality is the number of categories within a feature. In this dataset, each categorical feature would have held several categories, for example artists, title, movement etc.
- The small size of the dataset may not have allowed for significant learnings
- The inclusion of so few numerical features compared to categorical features may have been an issue
- Price cleaning may have some distorted values if dots were also used as decimal points. However, the chances of this are low because the dataset used dots as thousand separators.
- Lack of log transformation for skewed price distribution. This was initially attempted but presented challenges in consistency when trying to convert back from log price.

4.3. Conclusion

This initial iteration of the project provided valuable insights into the challenges of predicting art auction prices using machine learning. Several limitations were identified that will inform future improvements. First, the dataset exhibited high cardinality across multiple categorical features such as artist, title, and movement, which likely contributed to overfitting and poor generalization. Second, the small dataset size (754 records) restricted the model's ability to learn meaningful patterns and limited the robustness of evaluation metrics. Third, the imbalance between categorical and numerical features, with very few numeric variables, may have constrained the model's predictive power. These findings underscore the need for larger, more diverse datasets, feature engineering to reduce cardinality, and robust preprocessing strategies for numeric and categorical variables. Future iterations will also incorporate systematic log transformations, improved handling of categorical complexity, and additional features such as provenance, institutional recognition, and sentiment indicators. By addressing these limitations, subsequent versions of the model will aim for greater accuracy, stability, and practical utility in art price prediction.

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